

Question One: If $P(B) = 0.25$ and $P(A - B) = 0.5$ then we get

$$P(A|\Omega) - P(B)P(A|B) =$$

- A) 0.25 B) 0.5 C) 0 D) 0.2 E) None

Question Two: For some disjoint and independent events A, B with $P(A) =$
we have $P(B) =$

- A) 1 B) 0.3 C) 0.4 D) 0 E) None

Question Three: Let X be a discrete random variable with $E(X) = \bar{X}$

$$P(X = \bar{X}) = 0.25, \text{ then } P(X \neq \bar{X}) =$$

- A) 0 B) 0.8 C) 0.75 D) 0.2 E) None

Question Four: For some independent events A, B with $P(A)P(B) = 0.1$
we have $P(\bar{A} \cup \bar{B}) =$

- A) 0.9 B) 0.8 C) 0.7 D) 0.6 E) None

Question Five: Let A, B be two events with equal probability and
 $P(\bar{A} \cap B) + P(\bar{B} \cap A) = 0.2$, then $P(A) =$

- A) 0.2 B) 0.1 C) 0.8 D) 0.3 E) None

Question Six: Let X be a discrete random variable with $R_X = \{0, 1, -1\}$ then
have $E(X^2) + E(X^4) - 2E(X^6) =$

- A) -1 B) 1 C) 2 D) 0 E) None

Question Seven: Let f be a continuous strictly increasing function on $\mathbb{R}$. Then we have $\int_{-\infty}^{\infty} f(x)(\delta(x-1) - \delta(x+1)) dx$ is

- A) zero B) negative C) positive D) equal to one E) None

Question Eight: For the joint PDF $f(x, y) = 2x^b$ when $0 < x < 1$, $0 < y < 1$ and zero otherwise, the positive constant b equals

- A) 1 B) 0.5 C) 0 D) 2 E) None

Question Nine: For the joint PDF $f(x, y) = 0.5$ when $0 < x < 1$, $0 < y < 2$ and zero otherwise, we have $P(0 < Y < 1) =$

- A) 0.1 B) 0 C) 0.25 D) 0.5 E) None

Question Ten: A discrete random variable X with PDF

$P(X = x) = a(\delta(x) + \delta(x-1) + \delta(x+1) + \delta(x-2))$; where $a > 0$ is a constant, then we have $a + P(X = 0) + P(X = 1) =$

- A) 1 B) 0 C) 0.75 D) 0.5 E) None

Question Eleven ($\checkmark, X$): For continuous random variables X, Y we always have $\text{Cov}(-X-1, -Y) > \sigma_X \sigma_Y$ ~~X~~

Question Twelve ($\checkmark, X$): For continuous random variable X with CDF

$F(x) = 1 - e^{-\sqrt{x}}$ when $x > 0$ and zero otherwise; then the PDF is $f(x) = e^{-\sqrt{x}}/\sqrt{x}$ ~~X~~

Question Thirteen ($\checkmark, X$): If $x \in (-1, 1)$; then $u(-x-1) + \delta(x-2) = 0$ ~~X~~

Question Fourteen ($\checkmark, X$): For some random variable X , if the CDF

$F(x) = x^3, x \in (0, 1)$, then $P(X \in (0, 0.1)) + P(X = 0.5) = 0.001$ ~~X~~

Question Fifteen ($\checkmark, X$): Given the CDF

$$F(x) = \begin{cases} 0 & x < 1 \\ 0.25 & 1 \leq x < 2 \\ 1 & 2 \leq x \end{cases}$$

We have $P(X = 2) = 3/4$ ~~X~~



Al-Balqa Applied University

Faculty of Engineering Technology, Basic Science Department, MID Exam
Statistics and prob. for / Prob. and rand. processes, Summer 2021/2022

Student: _____

Instructor name: _____

Lecture Time: _____

Date: _____

Question Number	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Answer	B	A	E	D	A	B	C	D	B	C
Question Number	Q11	Q12	Q13	Q14	Q15					
Answer	B	A	E	D	C					

Q1: $P(\phi|\Omega) =$

- A) 0.1 B) 0 C) 0.3 D) 1 E) None

Q2: If A, B are independent, $P(A) = P(B)$ and $P(A \cup B) = 1$, then $P(A) =$

- A) 1 B) 0.3 C) 0 D) 0.5 E) None

Q3: If $P(X = x) = a\delta(x) + b\delta(x+1)$; then $a + b =$

- A) 0.5 B) 0.3 C) 0.4 D) 1 E) None

Q4: For $x \in (0, 1)$, we have $u(x) + 2u(1-x) + \delta(x) =$

- A) 0.2 B) 3 C) 1 D) 2 E) None

Q5: If $F(0) = 0$, and $0 < F(1) = \sqrt{0.5 P(0 < x < 1)}$, then $F(1) =$

- A) 0.5 B) 0.25 C) 0.4 D) 0.16 E) None

Q6: If $E(0.5 X^2 + X) = E(1 + X) = 0$ Then $Var(X) =$

- A) 0 B) 1 C) 2 D) 3 E) None

Q7: Let X, Y be two independent r.v's with $E(XY) = 3E(Y) = \int_{-\infty}^{\infty} \cos(x + \pi) \delta(x) dx$ then $E(X) =$

- A) 1 B) 2 C) 3 D) 4 E) None

Q8: For the joint PDF $f(x, y) = xy$ when $0 < x < a$, $0 < y < b$ and zero otherwise, we have ab equals

- A) 3 B) 1/2 C) 1/3 D) 2 E) None

Q9: If $\text{Corr}(-x, -y) = 0.25$ $\text{Cov}(X + 2, Y)/\sigma_Y$ then $\sigma_X =$

- A) 1 B) 2 C) 3 D) 4 E) none

Q10: For $X \sim B(n, 0.25)$ if $\text{Var}(X) = 3$, then $n =$

- A) 32 B) 4 C) 16 D) 2 E) none

Q11: For $X \sim B(n, p)$ if $\text{Var}(X) = 0.7 E(X)$, then $p =$

- A) 0.5 B) 0.3 C) 0.7 D) 1 E) None

Q12: If all points of $R_{XY} = \{(2, 2), (2, 3), (3, 2), (3, 3)\}$ have equal probability, then $f(2, 2) =$

- A) 0.25 B) 0.5 C) 0.75 D) 1 E) None

Q13: For the joint PDF $f(x, y) = b$ when $0 < x < 0.5$, $0 < y < 2$ and zero otherwise, we have $P(0 < x < 0.1 \text{ and } 0 < y < 0.1) =$

- A) 0.04 B) 0.2 C) 0.01 D) 0.1 E) None

Q14: If $E(\text{Var}(X) X) = E(X) E(X^2) - 8$, then $E(X) =$

- A) 0 B) 1 C) 3 D) 2 E) None

Q15: If A, B are independent events with $P(A) = P(B) = 0.8$ then we have $P(\bar{A} \cap \bar{B}) =$

- A) 0.01 B) 0.09 C) 0.04 D) 0.16 E) None